통계열역학 (2009년 1학기) 기말고사 답안

(총 140점)

1. **(총 25점, 각 5점)**

   (a) 1. 소정준 앙상블.

   (b) 4. 잔류 엔트로피.

   (c) 3. 등가중치 원리.

   (d) 7. para 수소.

   (e) 14. 요동.

2. **(총 30점, 각 10점)**

   (a)

$$\begin{aligned} p_1 &= \frac{e^{-\beta\epsilon_1}}{e^{-\beta\epsilon_1}+e^{-\beta\epsilon_2}} = \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}} \\ p_2 &= \frac{e^{-\beta\epsilon_2}}{e^{-\beta\epsilon_1}+e^{-\beta\epsilon_2}} = \frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}} \end{aligned}$$

   (b) i. $T=0$일 때,

$$\begin{aligned} \lim_{T\to 0} p_1 &= \lim_{T\to 0} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}} \\ &= \lim_{T\to 0} \frac{1}{1+e^{-(\epsilon_2-\epsilon_1)/k_BT}} \qquad (\text{양변을 } e^{-\epsilon_1/k_BT} \text{로 나눔}) \end{aligned}$$

   $\epsilon_2 > \epsilon_1$ 이므로

$$\lim_{T\to 0} \frac{1}{1+e^{-(\epsilon_2-\epsilon_1)/k_BT}} = 1$$

$$\begin{aligned} \lim_{T\to 0} p_2 &= \lim_{T\to 0} \frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_2/k_BT}+e^{-\epsilon_2/k_BT}} \\ &= \lim_{T\to 0} \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT}+1} \qquad (\text{양변을 } e^{-\epsilon_2/k_BT} \text{로 나눔}) \end{aligned}$$

   $\epsilon_2 > \epsilon_1$ 이므로

$$\lim_{T\to 0} \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT}+1} = 0$$

   ii. $T=\infty$ 일 때,

$$\lim_{T\to \infty} p_1 = \lim_{T\to \infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}}$$

$\lim_{T\to\infty} e^{-\epsilon_1/k_BT} = 1$ 그리고 $\lim_{T\to\infty} e^{-\epsilon_2/k_BT} = 1$ 이므로

$$\lim_{T\to\infty} p_1 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} = \frac{1}{2}$$

$$\lim_{T\to\infty} p_2 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}}$$

$\lim_{T\to\infty} e^{-\epsilon_1/k_BT} = 1$ 그리고 $\lim_{T\to\infty} e^{-\epsilon_2/k_BT} = 1$ 이므로

$$\lim_{T\to\infty} p_2 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} = \frac{1}{2}$$

(c) (a)에서

$$\begin{aligned} p_1 &= \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} \\ &= \frac{1}{1 + e^{-(\epsilon_2-\epsilon_1)/k_BT}} \qquad (\text{양변을 } e^{-\epsilon_1/k_BT} \text{로 나눔}) \\ &= \frac{1}{1 + e^{-\Delta\epsilon/k_BT}} \qquad (\Delta\epsilon = \epsilon_2 - \epsilon_1 \text{ 이므로}) \\ &= \frac{1}{1 + e^{-1}} = 0.731 \qquad (\Delta\epsilon = k_BT \text{ 이므로}) \end{aligned}$$

$$\begin{aligned} p_2 &= \frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} \\ &= \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT} + 1} \qquad (\text{양변을 } e^{-\epsilon_2/k_BT} \text{로 나눔}) \\ &= \frac{1}{e^{\Delta\epsilon/k_BT} + 1} \qquad (\Delta\epsilon = \epsilon_2 - \epsilon_1 \text{ 이므로}) \\ &= \frac{1}{e^{1} + 1} = 0.269 \qquad (\Delta\epsilon = k_BT \text{ 이므로}) \end{aligned}$$

3. **(총 20점, 각 10점)**

(a) 이원자분자의 진동 에너지 준위:

$$\epsilon_n = \left(n + \frac{1}{2}\right)\hbar\omega = \left(n + \frac{1}{2}\right)h\nu = \left(n + \frac{1}{2}\right)hc\bar{\nu} \qquad (n = 0, 1, 2, 3, \cdots) \qquad (1)$$

위 식에서 $\omega = 2\pi\nu = (k/\mu)^{1/2}$.

$$\begin{aligned} q_{vib}(T) = \sum_{n=1} e^{-\beta\epsilon_n} &= e^{-\beta h\nu/2} \sum_{n=1} e^{-\beta h\nu n} = \frac{e^{-\beta h\nu/2}}{1 - e^{-\beta h\nu}}, \\ &= \frac{e^{-\Theta_{vib}/2T}}{1 - e^{-\Theta_{vib}/T}}. \end{aligned} \qquad (2)$$

위 식에서 Θ_{vib} 는 진동 온도 (vibrational temperature) 이며 다음과 같이 정의된다:

$$\Theta_{vib} = \frac{h\nu}{k_B} = \frac{hc\bar{\nu}}{k_B}. \tag{3}$$

(b) 이원자분자가 진동 바닥상태에 존재할 확률:

$$\begin{aligned} p_{n=0}(T) &= \frac{e^{-\beta\epsilon_0}}{q} \\ &= \frac{e^{-\beta h\nu/2}}{e^{-\beta h\nu/2}/(1-e^{-\beta h\nu})} \\ &= 1 - e^{-\beta h\nu} \\ &= 1 - e^{-h\nu/k_BT} = 1 - e^{-\Theta_{vib}/T} \\ &= 1 - e^{-2274\ \mathrm{K}/5000\ \mathrm{K}} = 0.365 \end{aligned}$$

이원자분자가 진동 첫번째 들뜬상태에 존재할 확률:

$$\begin{aligned} p_{n=1}(T) &= \frac{e^{-\beta\epsilon_1}}{q} \\ &= \frac{e^{-3\beta h\nu/2}}{e^{-\beta h\nu/2}/(1-e^{-\beta h\nu})} \\ &= e^{-\beta h\nu}(1 - e^{-\beta h\nu}) \\ &= e^{-h\nu/k_BT}(1 - e^{-h\nu/k_BT}) = e^{-\Theta_{vib}/T}(1 - e^{-\Theta_{vib}/T}) \\ &= e^{-2274\ \mathrm{K}/5000\ \mathrm{K}}(1 - e^{-2274\ \mathrm{K}/5000\ \mathrm{K}}) = 0.231 \end{aligned}$$

4. **(총 20점, 각 10점)**

(a)

$$\begin{aligned} \langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q(N,V,T)}{\partial T}\right)_{N,V}, \\ &= k_BT^2\frac{\partial}{\partial T}\left(\frac{q_1^{N_1}q_2^{N_2}}{N_1!N_2!}\right), \\ &= k_BT^2\frac{\partial}{\partial T}\Big(N_1\ln q_1 + N_2\ln q_2 - \ln N_1! - \ln N_2!\Big), \\ &= k_BT^2\frac{\partial}{\partial T}\left[\frac{3N_1}{2}\ln T + \frac{3N_1}{2}\ln\left(\frac{2\pi m_1k_B}{h^2}\right) + N_1\ln V - \ln N_1!\right] \\ &\quad + k_BT^2\frac{\partial}{\partial T}\left[\frac{3N_2}{2}\ln T + \frac{3N_2}{2}\ln\left(\frac{2\pi m_2k_B}{h^2}\right) + N_2\ln V - \ln N_2!\right] \\ &= \frac{3}{2}N_1k_BT + \frac{3}{2}N_2k_BT. \end{aligned}$$

(b)

$$
\begin{aligned}
P &= -\left(\frac{\partial A}{\partial V}\right)_{N,T}, \\
&= k_B T\left(\frac{\partial \ln Q}{\partial V}\right)_{N,T}, \\
&= k_B T \frac{\partial}{\partial V}\left(\frac{q_1^{N_1} q_2^{N_2}}{N_1! N_2!}\right), \\
&= k_B T \frac{\partial}{\partial V}\Big(N_1 \ln q_1 + N_2 \ln q_2 - \ln N_1! - \ln N_2!\Big), \\
&= k_B T \frac{\partial}{\partial V}\left\{N_1 \ln\left(\left(\frac{2\pi m_1 k_B T}{h^2}\right)^{3/2} V\right) - \ln N_1!\right\} \\
&\quad + k_B T \frac{\partial}{\partial V}\left\{N_2 \ln\left(\left(\frac{2\pi m_2 k_B T}{h^2}\right)^{3/2} V\right) - \ln N_2!\right\}, \\
&= N_1 k_B T \frac{\partial}{\partial V}\left\{\ln\left(\frac{2\pi m_1 k_B T}{h^2}\right)^{3/2} + \ln V - \ln N_1!\right\}, \\
&\quad + N_2 k_B T \frac{\partial}{\partial V}\left\{\ln\left(\frac{2\pi m_2 k_B T}{h^2}\right)^{3/2} + \ln V - \ln N_2!\right\}, \\
&= \frac{N_1 k_B T}{V} + \frac{N_2 k_B T}{V} \Rightarrow PV = (N_1 + N_2) k_B T
\end{aligned}
$$

5. **(총 25점)**

(a) **(10점)** 바닥상태의 에너지는 0이고 첫번째 들뜬상태의 에너지는 300 cm^{-1} 이므로

$$
\begin{aligned}
&p_{e1} = 0.15 = \frac{e^{-\beta\epsilon_{e1}}}{1 + e^{-\beta\epsilon_{e1}}} = \frac{e^{-\epsilon_{e1}/k_B T}}{1 + e^{-\epsilon_{e1}/k_B T}} = \frac{e^{-(300\ \text{cm}^{-1})hc/k_B T}}{1 + e^{-(300\ \text{cm}^{-1})hc/k_B T}} \\
&0.15 + 0.15 e^{-(300\ \text{cm}^{-1})hc/k_B T} = e^{-(300\ \text{cm}^{-1})hc/k_B T} \\
&0.15 = 0.85 e^{-(300\ \text{cm}^{-1})hc/k_B T} \\
&0.176 = e^{-(300\ \text{cm}^{-1})hc/k_B T} \\
&\ln 0.176 = -\frac{(300\ \text{cm}^{-1})hc}{k_B T} \Rightarrow -1.74 = -\frac{(300\ \text{cm}^{-1})hc}{k_B T} \\
&T = \frac{(300\ \text{cm}^{-1})hc}{1.74 \times k_B} = \frac{(300\ \text{cm}^{-1})(6.62608 \times 10^{-34}\ \text{J s})(2.998 \times 10^{10}\text{cm s}^{-1})}{(1.74)(1.38066 \times 10^{-23}\ \text{J K}^{-1})} = 248\ \text{K}
\end{aligned}
$$

(b) **(10점)** 바닥상태의 에너지는 0이고 첫번째 들뜬상태의 에너지는 300 cm^{-1} 이고 들뜬

상태의 축퇴도가 2이므로

$$p_{e1} = 0.15 = \frac{2e^{-\beta\epsilon_{e1}}}{1+2e^{-\beta\epsilon_{e1}}} = \frac{2e^{-\epsilon_{e1}/k_BT}}{1+2e^{-\epsilon_{e1}/k_BT}} = \frac{2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}}{1+2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}}$$

$$0.15 + 0.30e^{-(300\ \mathrm{cm}^{-1})hc/k_BT} = 2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$0.15 = 1.70e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$0.088 = e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$\ln 0.088 = -\frac{(300\ \mathrm{cm}^{-1})hc}{k_BT} \Rightarrow -2.43 = -\frac{(300\ \mathrm{cm}^{-1})hc}{k_BT}$$

$$T = \frac{(300\ \mathrm{cm}^{-1})hc}{1.74 \times k_B} = \frac{(300\ \mathrm{cm}^{-1})(6.62608\times10^{-34}\ \mathrm{J\ s})(2.998\times10^{10}\mathrm{cm\ s}^{-1})}{(2.43)(1.38066\times10^{-23}\ \mathrm{J\ K}^{-1})} = 178\ \mathrm{K}$$

(c) **(5점)** (b)에서 얻은 온도가 (a)에서 얻은 온도보다 낮다. 그 이유는 (b)에는 들뜬 에너지준위에 2개의 미시상태가 존재하므로 계가 들뜬 에너지준위에 존재할 확률이 증가하기 때문이다.

6. **(총 20점)**

(a) 300 K에서

$$\begin{aligned} q_{elec}(300\ \mathrm{K}) &\approx 4 + 2e^{-875.4\ \mathrm{cm}^{-1}hc/k_BT} \\ &= 4 + 2\times\frac{(875.4\ \mathrm{cm}^{-1})(6.62608\times10^{-34}\ \mathrm{J\ s})(2.998\times10^{10}\mathrm{cm\ s}^{-1})}{(1.38066\times10^{-23}\ \mathrm{J\ K}^{-1})(300\ \mathrm{K})} = 4.030 \end{aligned}$$

(b) 1000 K에서

$$\begin{aligned} q_{elec}(1000\ \mathrm{K}) &\approx 4 + 2e^{-875.4\ \mathrm{cm}^{-1}hc/k_BT} \\ &= 4 + 2\times\frac{(875.4\ \mathrm{cm}^{-1})(6.62608\times10^{-34}\ \mathrm{J\ s})(2.998\times10^{10}\mathrm{cm\ s}^{-1})}{(1.38066\times10^{-23}\ \mathrm{J\ K}^{-1})(1000\ \mathrm{K})} = 4.568 \end{aligned}$$